

CHAPTER II.2, WEEK 2

Noah Parker

2.1. Let A be a ring, $X = \operatorname{Spec} A$ its spectrum. Fix an element $f \in A$, $D(f) = V(f)^c \subset X$ the associated distinguished open set. Show that $(D(f), \mathcal{O}_X|_{D(f)}) \cong \operatorname{Spec} A_f$ as locally ringed spaces.

Proof. Let $Y = \operatorname{Spec} A_f$, $S = \{f^n\}_{n \geq 1}$, and $\phi : A \rightarrow S^{-1}A = A_f$ be the natural map $a \mapsto a/1$. Note that $f^n \in \mathfrak{p}$ prime for some $n \geq 1$ iff $f \in \mathfrak{p}$ —that is, $S \cap \mathfrak{p} = \emptyset$ iff $\mathfrak{p} \notin D(f)$. Thus, AM 3.11.iv gives us a bijection

$$\begin{aligned} D(f) &\longleftrightarrow Y \\ \mathfrak{p} &\longrightarrow S^{-1}\mathfrak{p}, \\ \phi^*(\mathfrak{q}) &\longleftarrow \mathfrak{q}. \end{aligned}$$

Clearly the map $\mathfrak{q} \mapsto \phi^*(\mathfrak{q})$ is continuous, so it suffices to show that $\mathfrak{p} \mapsto S^{-1}\mathfrak{p}$ is continuous.

Let $V(\mathfrak{b}) \subset A_f$ be closed, and denote $\psi : \mathfrak{p} \mapsto S^{-1}\mathfrak{p}$. AM 3.11.i gives us that $\mathfrak{b} = S^{-1}\mathfrak{a}$ is extended. Then $\mathfrak{p} \supset \mathfrak{a}$ implies $S^{-1}\mathfrak{p} \supset S^{-1}\mathfrak{a} = \mathfrak{b}$ whenever $f \notin \mathfrak{a}, \mathfrak{p}$, so that $V(\mathfrak{a}) \cap D(f) \subset \psi^{-1}(\mathfrak{b})$. Suppose that $S^{-1}\mathfrak{p} \supset S^{-1}\mathfrak{a}$. That is, for all $a \in \mathfrak{a}$, there is some $x \in \mathfrak{p}$ and $n \geq 1$ such that $f^n(a - x) = 0$. Hence, $f^n a \in \mathfrak{p}$, so that $\mathfrak{p}$ prime and $f \notin \mathfrak{p}$ gives us $a \in \mathfrak{p}$. We conclude that $\psi^{-1}(V(\mathfrak{b})) = V(\mathfrak{a}) \cap D(f) \subset D(f)$ is closed, so that $\psi = (\psi^*)^{-1}$ is continuous.

Note that $(\mathcal{O}_X|_{D(f)})_{\mathfrak{p}} \cong A_{\mathfrak{p}} \cong (A_{\mathfrak{p}})_f \cong (A_f)_{\psi(\mathfrak{p})} \cong (\mathcal{O}_Y)_{\psi(\mathfrak{p})}$ for each $\mathfrak{p} \notin f$. Call this isomorphism $\alpha_{\mathfrak{p}} : (\mathcal{O}_X|_{D(f)})_{\mathfrak{p}} \rightarrow (\mathcal{O}_Y)_{\psi(\mathfrak{p})}$, and let

$$\alpha : \sqcup_{\mathfrak{p} \in D(f)} (\mathcal{O}_X|_{D(f)})_{\mathfrak{p}} \rightarrow \sqcup_{\mathfrak{q} \in Y} (\mathcal{O}_Y)_{\mathfrak{q}}$$

be defined by $\alpha|_{(\mathcal{O}_X|_{D(f)})_{\mathfrak{p}}} = \alpha_{\mathfrak{p}}$. Then we have a map

$$\begin{aligned} \phi^{\sharp}(U) : \mathcal{O}_X|_{D(f)}(U) &\rightarrow (\phi^*)_* \mathcal{O}_Y(U) \\ s &\mapsto \alpha \circ s \circ \phi^*. \end{aligned}$$

We have that

$$\alpha \circ s|_V \circ \phi^* = \alpha \circ s \circ \phi^*|_{(\phi^*)^{-1}(V)}$$

as functions on $(\phi^*)^{-1}(V)$ whenever $V \subset U \subset D(f)$ are open and $s \in \mathcal{O}_X(U)$, so $\phi^{\sharp}(U)$ gives a sheaf homomorphism $\phi^{\sharp} : \mathcal{O}_X|_{D(f)} \rightarrow \mathcal{O}_Y$. We have that $\phi^{\sharp}_{\mathfrak{p}} = \alpha_{\mathfrak{p}}$ is an isomorphism of rings for all $\mathfrak{p} \in D(f)$, so $\phi^{\sharp}$ is itself a sheaf isomorphism. Then $(f, \phi^{\sharp})$ is an isomorphism of locally ringed spaces, since ring isomorphisms preserve maximality. $\square$

2.2. Let $(X, \mathcal{O}_X)$ be a scheme, and let $U \subset X$ be any open subset. Show that $(U, \mathcal{O}_X|_U)$ is a scheme. We call this the *induced scheme structure* on the open set U , and we refer to $(U, \mathcal{O}_X|_U)$ as the open subscheme of U .

Proof. Choose a basic cover $\{D(f_i)\}$ of U . Then each

$$(D(f_i), (\mathcal{O}_X|_U)_{D(f_i)}) = (D(f_i), \mathcal{O}_X|_{D(f_i)}) \cong \operatorname{Spec} A_{f_i}$$

by 2.1. We conclude that $\{D(f_i)\}$ is an affine open cover of $(U, \mathcal{O}_X|_U)$, so that the latter locally ringed space is a sheaf. $\square$

- 2.3. (*) A scheme $(X, \mathcal{O}_X)$ is *reduced* if for every open set $U \subset X$, the ring $\mathcal{O}_X(U)$ has no nilpotent elements.
- (a) Show that $(X, \mathcal{O}_X)$ is reduced if and only if for every $p \in X$, the local ring $\mathcal{O}_{X,p}$ has no nilpotent elements.
 - (b) Let $(X, \mathcal{O}_X)$ be a scheme. Let $(\mathcal{O}_X)_{\text{red}}$ be the sheaf associated to the presheaf $U \mapsto \mathcal{O}_X(U)_{\text{red}}$, where for any ring A , we denote by A_{red} the quotient of A by its nilradical $\mathfrak{N}(A)$. Show that $(X, (\mathcal{O}_X)_{\text{red}})$ is a scheme. We call it the *reduced scheme* associated to X , and denote it by X_{red} . Show that there is a morphism of schemes $X_{\text{red}} \rightarrow X$, which is a homeomorphism on the underlying topological spaces.
 - (c) Let $f : X \rightarrow Y$ be a morphism of schemes, and assume that X is reduced. Show that there is a unique morphism $g : X \rightarrow Y_{\text{red}}$ such that f is obtained by composing g with the natural map $Y_{\text{red}} \rightarrow Y$.
- 2.4. Let A be a ring, and let $(X, \mathcal{O}_X)$ be a scheme. Given a morphism $f : X \rightarrow \text{Spec } A$, we have an associated map $f^\sharp : \mathcal{O}_{\text{Spec } A} \rightarrow f_* \mathcal{O}_X$ on sheaves. Taking Global sections, we obtain a homomorphism $A \rightarrow \Gamma(X, \mathcal{O}_X)$. Thus, there is a natural map

$$\alpha : \text{Hom}_{\mathfrak{S}ch}(X, \text{Spec } A) \rightarrow \text{Hom}_{\mathfrak{Ring}}(A, \Gamma(X, \mathcal{O}_X)).$$

Show that α is bijective.

Proof. Suppose $f, g : X \rightarrow \text{Spec } A$ are two morphisms of schemes with $\alpha(f) = \alpha(g)$. That is, $f^\sharp(\text{Spec } A) \equiv g^\sharp(\text{Spec } A)$. Fix an affine cover $\{U_i \cong \text{Spec } B_i\}$ of X □

- 2.5. Describe $\text{Spec } \mathbb{Z}$, and show that it is a final object for the category of schemes, i.e., each scheme X admits a unique morphism to $\text{Spec } \mathbb{Z}$.

Proof. The only primes of $\mathbb{Z}$ are (p) for p prime, and (0) . The former are all maximal, while the latter is a generic point for $\text{Spec } \mathbb{Z}$, since $V(\mathfrak{a}) \ni (0)$ implies $(0) \supset \mathfrak{a}$, i.e. $\mathfrak{a} = (0) \subset \mathfrak{p}$ for all $\mathfrak{p}$ prime.

We have a bijection

$$\text{Hom}_{\mathfrak{S}ch}(X, \text{Spec } \mathbb{Z}) \rightarrow \text{Hom}_{\mathfrak{Ring}}(\mathbb{Z}, \Gamma(X, \mathcal{O}_X)),$$

so that the former is singleton for all X by the fact that $\mathbb{Z}$ is initial in $\mathfrak{Ring}$. □

- 2.7. Let X be a scheme. For any point $x \in X$, let $\mathcal{O}_x$ be the local ring at x , and m_x its maximal ideal. We define the *residue field* of x on X to be the field $k(x) = \mathcal{O}_x/m_x$. Now, let K be any field. Show that to give a morphism of $\text{Spec } K$ to X , it is equivalent to give a point $x \in X$ and an inclusion map $k(x) \rightarrow K$.

Proof. □

- 2.8. (*) Let X be a scheme. For any point $x \in X$, we define the *Zariski tangent space* T_x at $x \in X$ to be the dual of the $k(x)$ -vector space m_x/m_x^2 . Now, assume that X is a scheme over a field k , and let $k[\epsilon]/\epsilon^2$ be the *ring of dual numbers* over k . Show that to give a k -morphism of $\text{Spec } k[\epsilon]/\epsilon^2$ to X is equivalent to giving a point $x \in X$, *rational over* k (i.e. such that $k(x) = k$), and an element of T_x .
- 2.9. If X is a topological space, and Z an irreducible closed subset of X , a generic point for Z is a point ζ such that $Z = \{\zeta\}^-$. If X is a scheme, show that every (nonempty) irreducible closed subset has a unique generic point.

Proof. Let Z be an irreducible closed subset of a scheme X , and let $\{U_i \cong \operatorname{Spec} A_i\}$ be an affine open cover of X . Then $\square$

2.10. Describe $\operatorname{Spec} \mathbb{R}[x]$. How does its topological space compare to the set $\mathbb{R}$? To $\mathbb{C}$?

Proof. $\square$

2.13. A topological space is *quasi-compact* if every open cover has a finite sub-cover.

- (a) Show that a topological space X is noetherian if and only if every open subset is quasi-compact.
- (b) If A is a noetherian ring, show that $(\operatorname{Spec} A)$ is a noetherian topological space.
- (c) Give an example to show that $(\operatorname{Spec} A)$ can be noetherian even when A is not.